Phase 2: Solution Requirements Specification

Functional & Non-Functional System Requirements

1. Functional Requirements

• FR-01: Data Cleaning - Validate schema, handle data types, and document constant raw health_risk_score.

• FR-02: Risk Score Formulation - Implement reproducible calibrated composite score (0-100) across 8 sub-factors.

• FR-03: Visual Analytics - Deliver 8 publication-grade charts, 2 Tableau dashboards, and 1 multi-scene story.

• FR-04: Web Application - Provide responsive Flask portal with REST API, patient explorer, and risk calculator.

2. Non-Functional Requirements

• NFR-01: Performance - API responses must execute in < 250ms for cohort queries.

• NFR-02: Integrity - Zero fabricated findings; 100% reproducible statistical outputs.

• NFR-03: Test Coverage - Automated pytest test suite covering data, routes, and workbook XML.